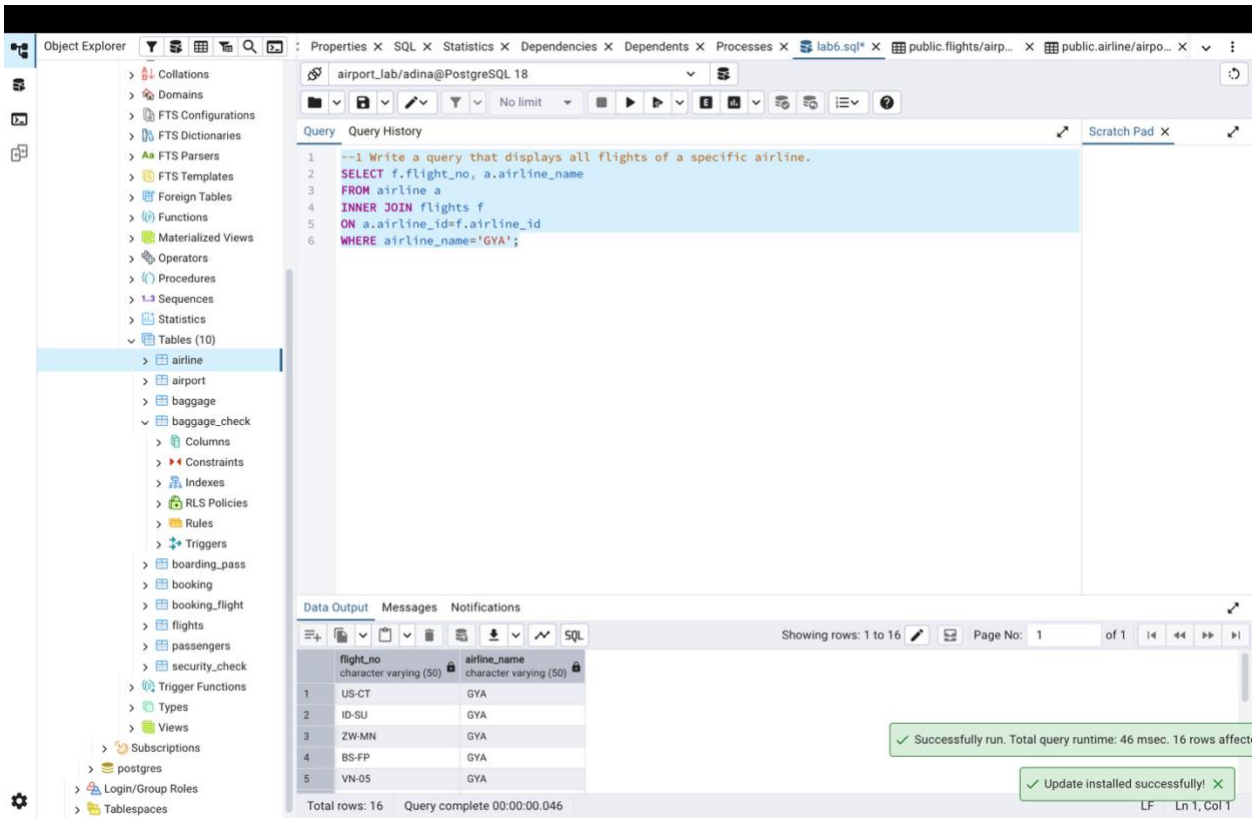
Laboratory work 6

Submission format: word or pdf document containing screenshots of your tasks with answers that covers full-page (Alt+PrtSc).

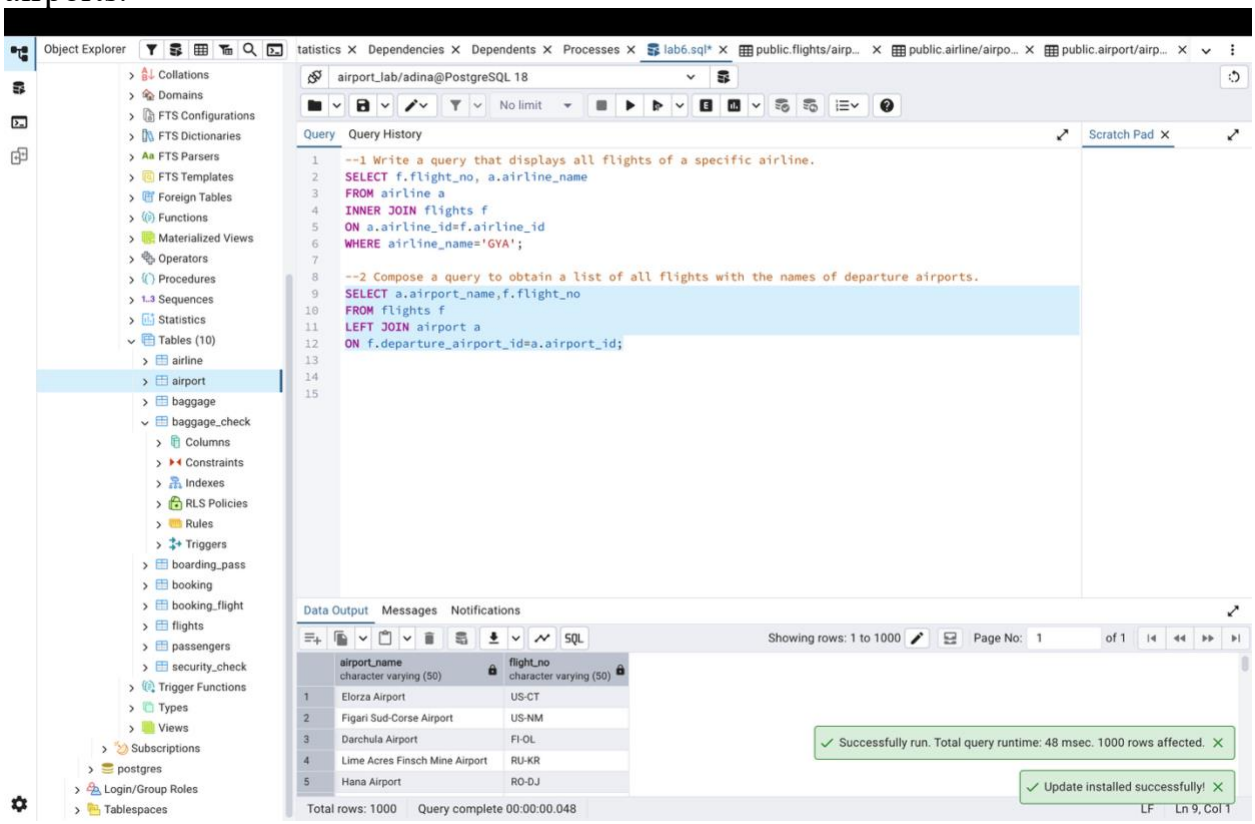
JOIN

1. Write a query that displays all flights of a specific airline.



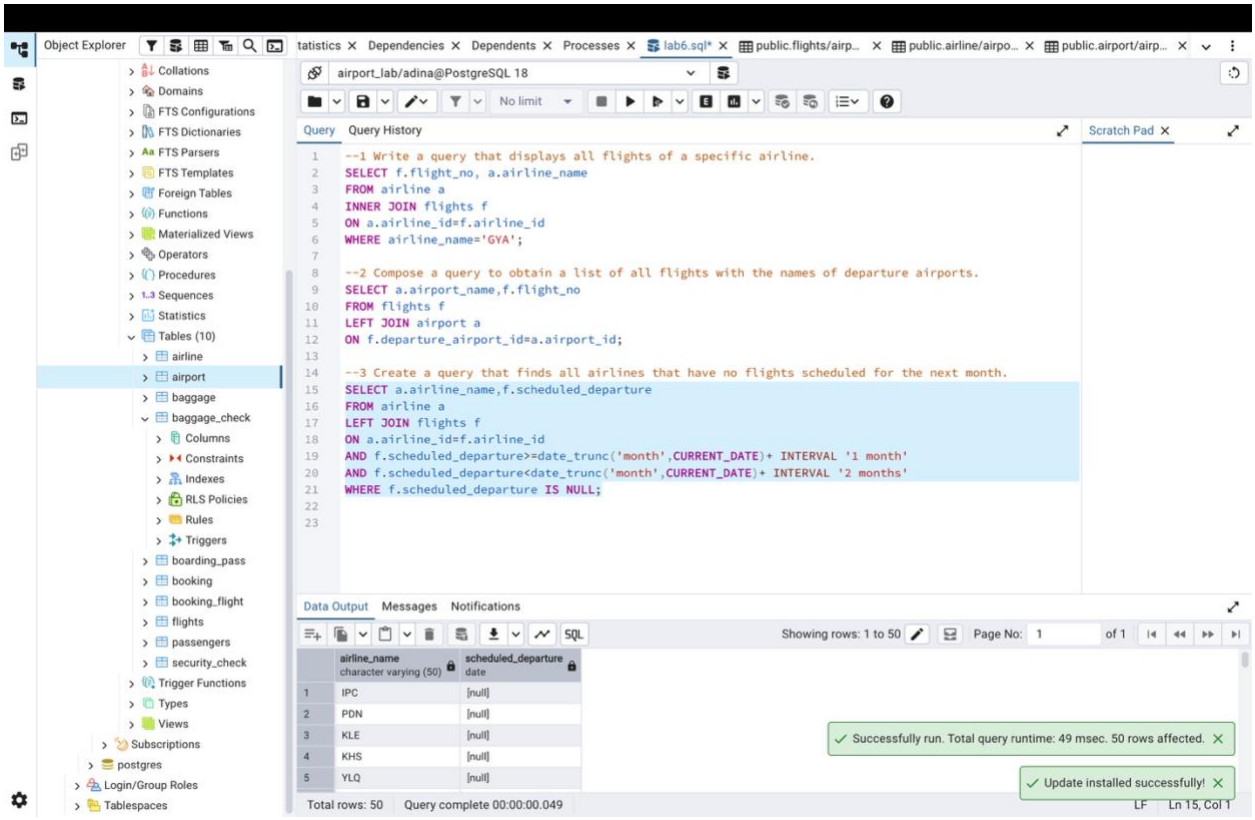
2. Compose a query to obtain a list of all flights with the names of departure

airports.

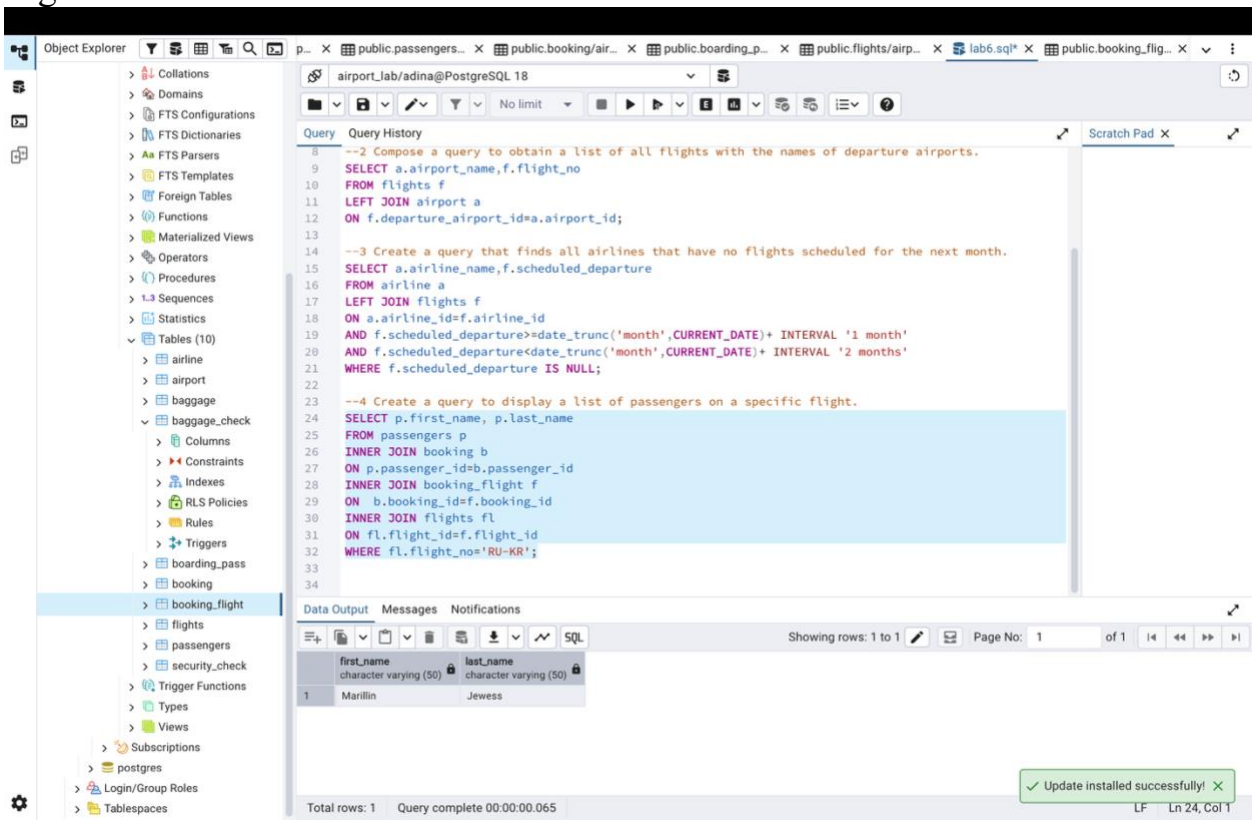


3. Create a query that finds all airlines that have no flights scheduled for the next

month.

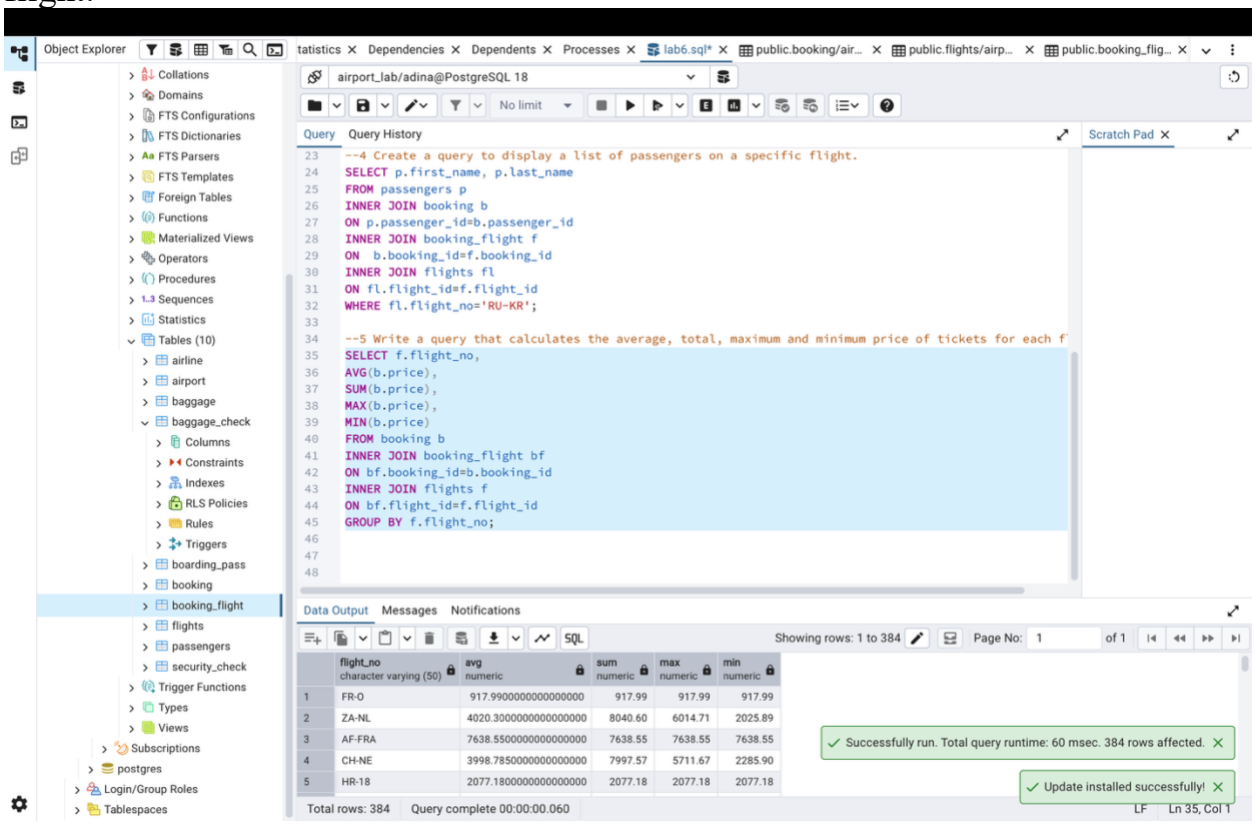

Object Explorer
tatistics
Dependencies
Dependents
Processes
lab6.sql*
public.flights/airp...
public.airline/airpo...
public.airport/airp...
airport_lab/adina@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
--1 Write a query that displays all flights of a specific airline.
SELECT f.flight_no, a.airline_name
FROM airline a
INNER JOIN flights f
ON a.airline_id=f.airline_id
WHERE airline_name='GYA';
--2 Compose a query to obtain a list of all flights with the names of departure airports.
SELECT a.airport_name,f.flight_no
FROM flights f
LEFT JOIN airport a
ON f.departure_airport_id=a.airport_id;
--3 Create a query that finds all airlines that have no flights scheduled for the next month.
SELECT a.airline_name,f.scheduled_departure
FROM airline a
LEFT JOIN flights f
ON a.airline_id=f.airline_id
AND f.scheduled_departure>=date_trunc('month',CURRENT_DATE)+ INTERVAL '1 month'
AND f.scheduled_departure<date_trunc('month',CURRENT_DATE)+ INTERVAL '2 months'
WHERE f.scheduled_departure IS NULL;
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Statistics
Tables (10)
airline
airport
baggage
baggage_check
Columns
Constraints
Indexes
RLS Policies
Rules
Triggers
boarding_pass
booking
booking_flight
flights
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
postgres
Login/Group Roles
Tablespaces
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 50
Page No: 1
of 1
airline_name
character varying (50)
scheduled_departure
date
1 IPC [null]
2 PDN [null]
3 KLE [null]
4 KHS [null]
5 YLQ [null]
Successfully run. Total query runtime: 49 msec. 50 rows affected.
Update installed successfully!
Total rows: 50
Query complete 00:00:00.049
LF
Ln 15, Col 1

4. Create a query to display a list of passengers on a specific flight.



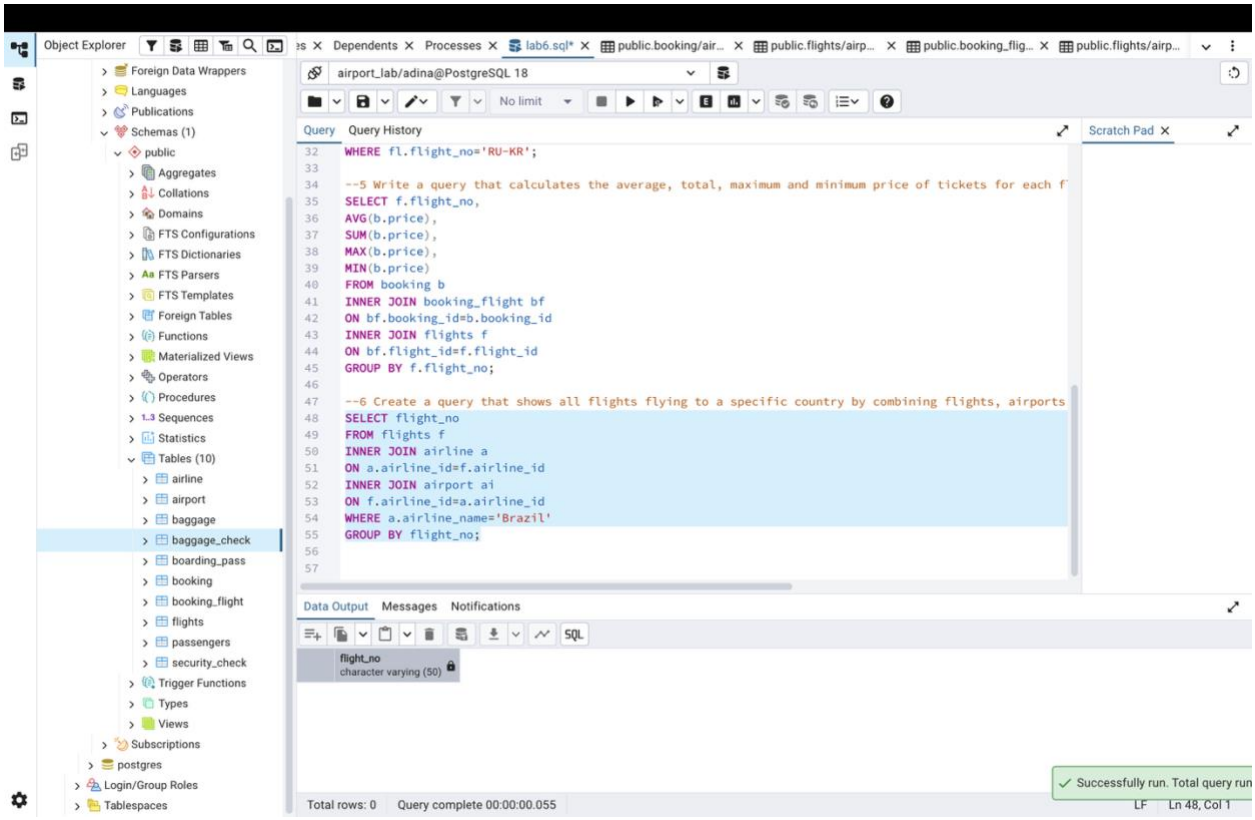
5. Write a query that calculates the average, total, maximum and minimum price of tickets for each

flight.

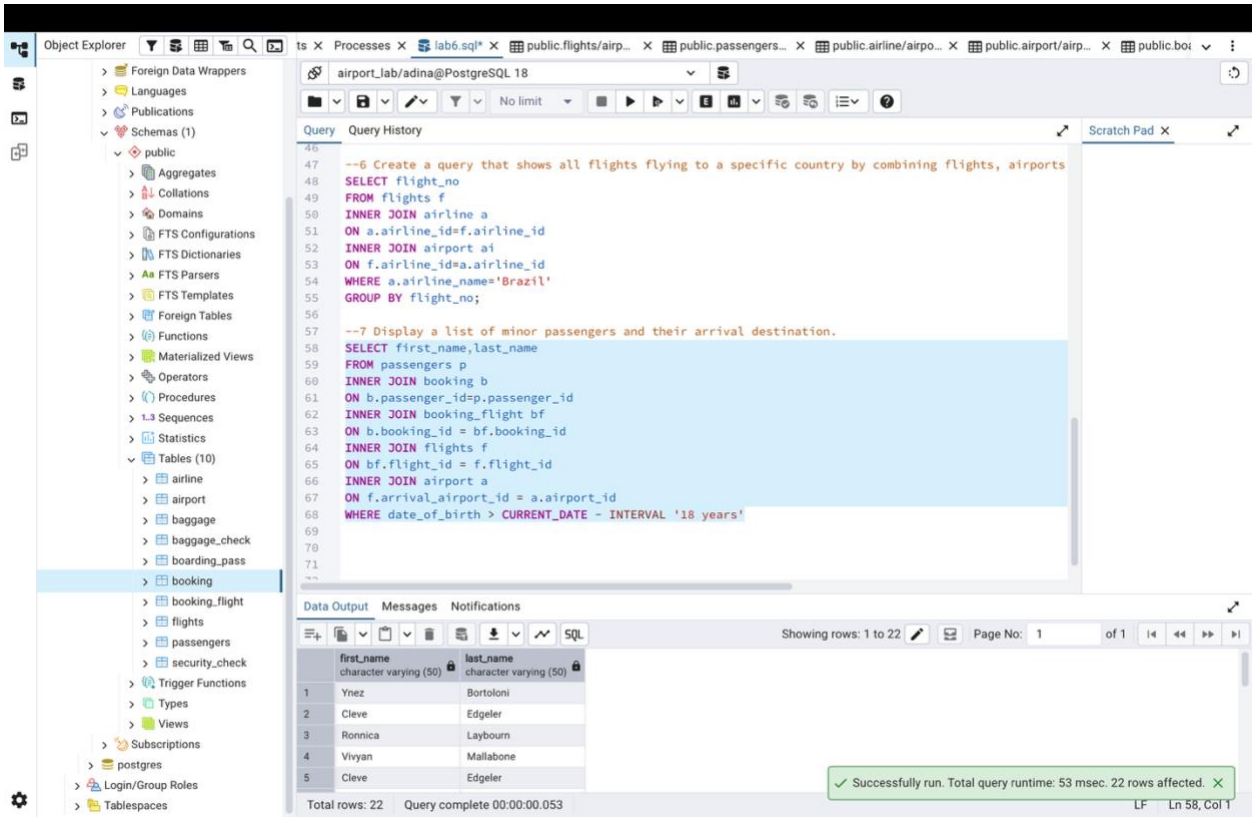


6. Create a query that shows all flights flying to a specific country by combining flights, airports and airline, and using the condition on the country

name.

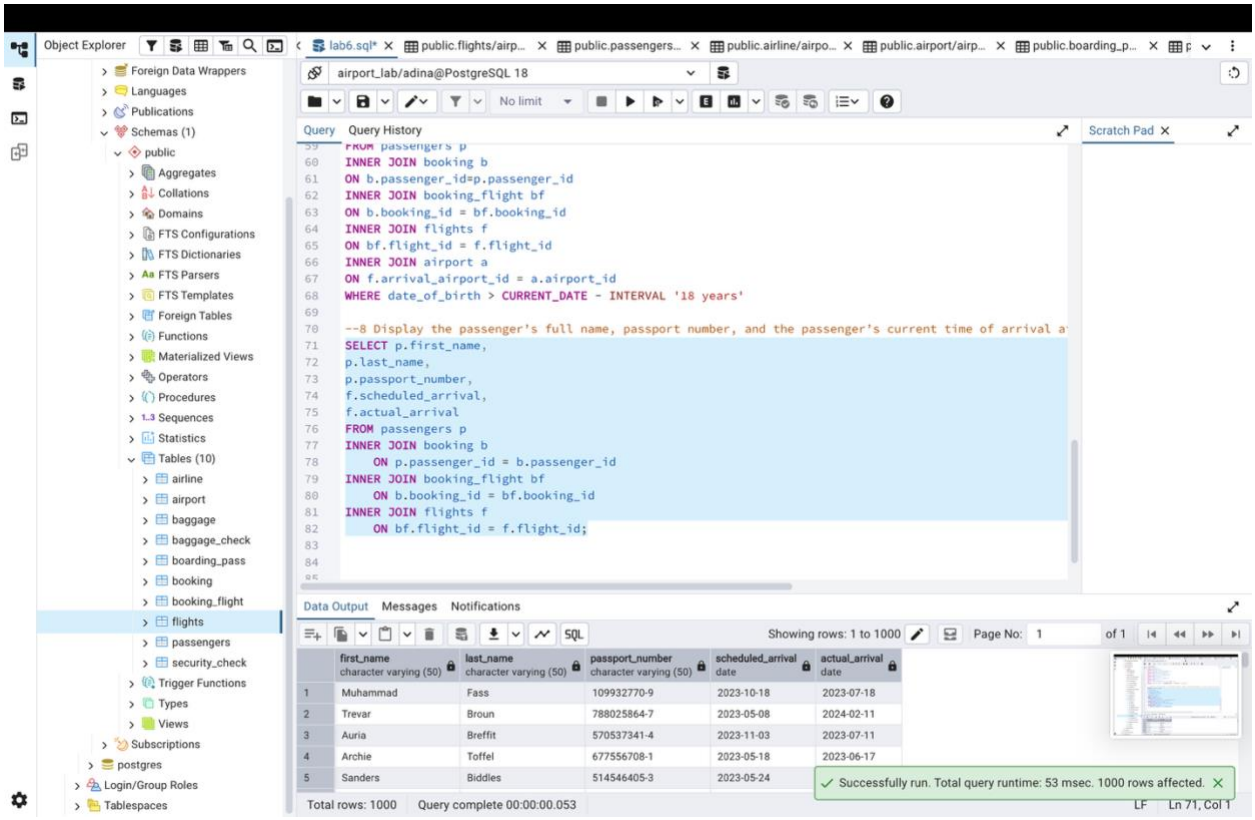

Object Explorer
Dependents
Processes
lab6.sql*
public.booking/air...
public.flights/airp...
public.booking_flig...
public.flights/airp...
airport_lab/adina@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
WHERE fl.flight_no='RU-KR';
--5 Write a query that calculates the average, total, maximum and minimum price of tickets for each f
SELECT f.flight_no,
AVG(b.price),
SUM(b.price),
MAX(b.price),
MIN(b.price)
FROM booking b
INNER JOIN booking_flight bf
ON bf.booking_id=b.booking_id
INNER JOIN flights f
ON bf.flight_id=f.flight_id
GROUP BY f.flight_no;
--6 Create a query that shows all flights flying to a specific country by combining flights, airports
SELECT flight_no
FROM flights f
INNER JOIN airline a
ON a.airline_id=f.airline_id
INNER JOIN airport ai
ON f.airline_id=a.airline_id
WHERE a.airline_name='Brazil'
GROUP BY flight_no;
Data Output
Messages
Notifications
SQL
flight_no
character varying (50)
Successfully run. Total query run
Total rows: 0
Query complete 00:00:00.055
LF
Ln 48, Col 1
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Statistics
Tables (10)
airline
airport
baggage
baggage_check
boarding_pass
booking
booking_flight
flights
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
postgres
Login/Group Roles
Tablespaces

7. Display a list of minor passengers and their arrival destination.



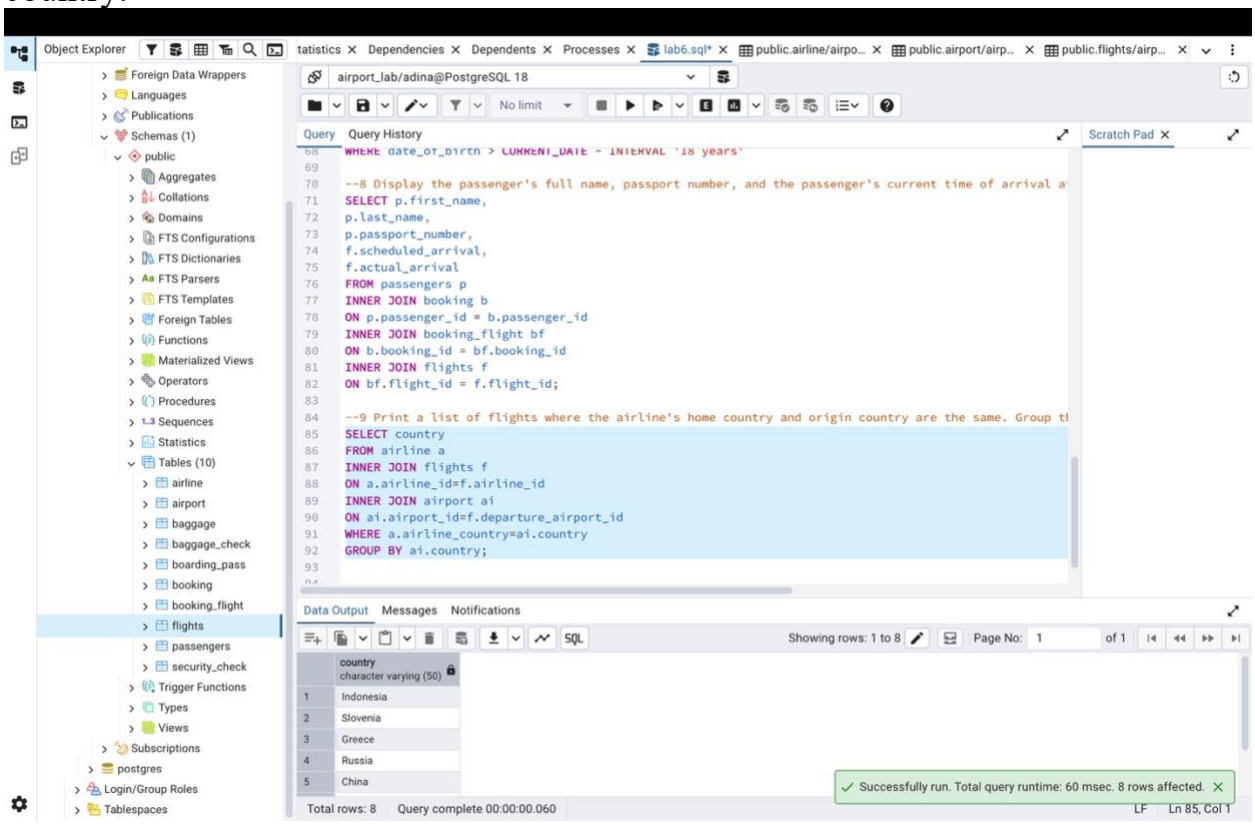
8. Display the passenger's full name, passport number, and the passenger's current time of arrival at the

destination.



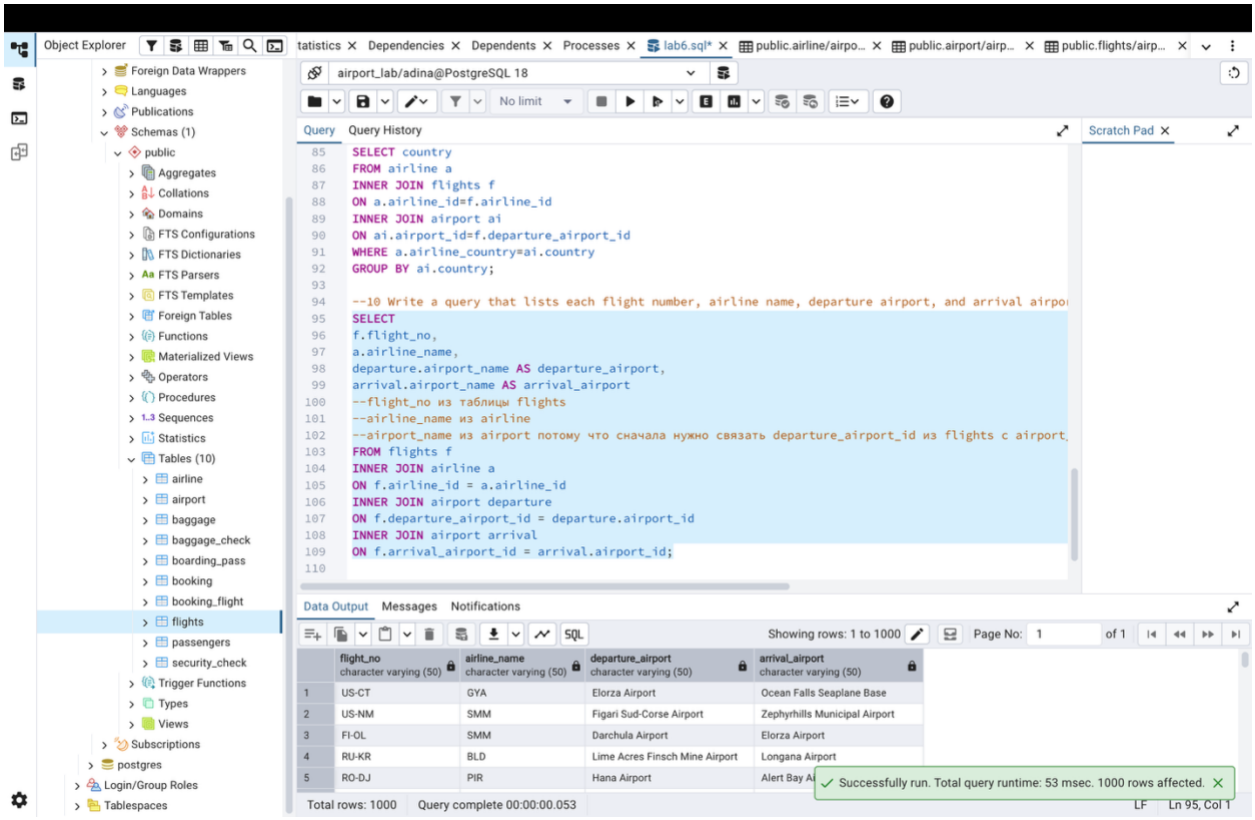
9. Print a list of flights where the airline's home country and origin country are the same. Group them by the airport

country.

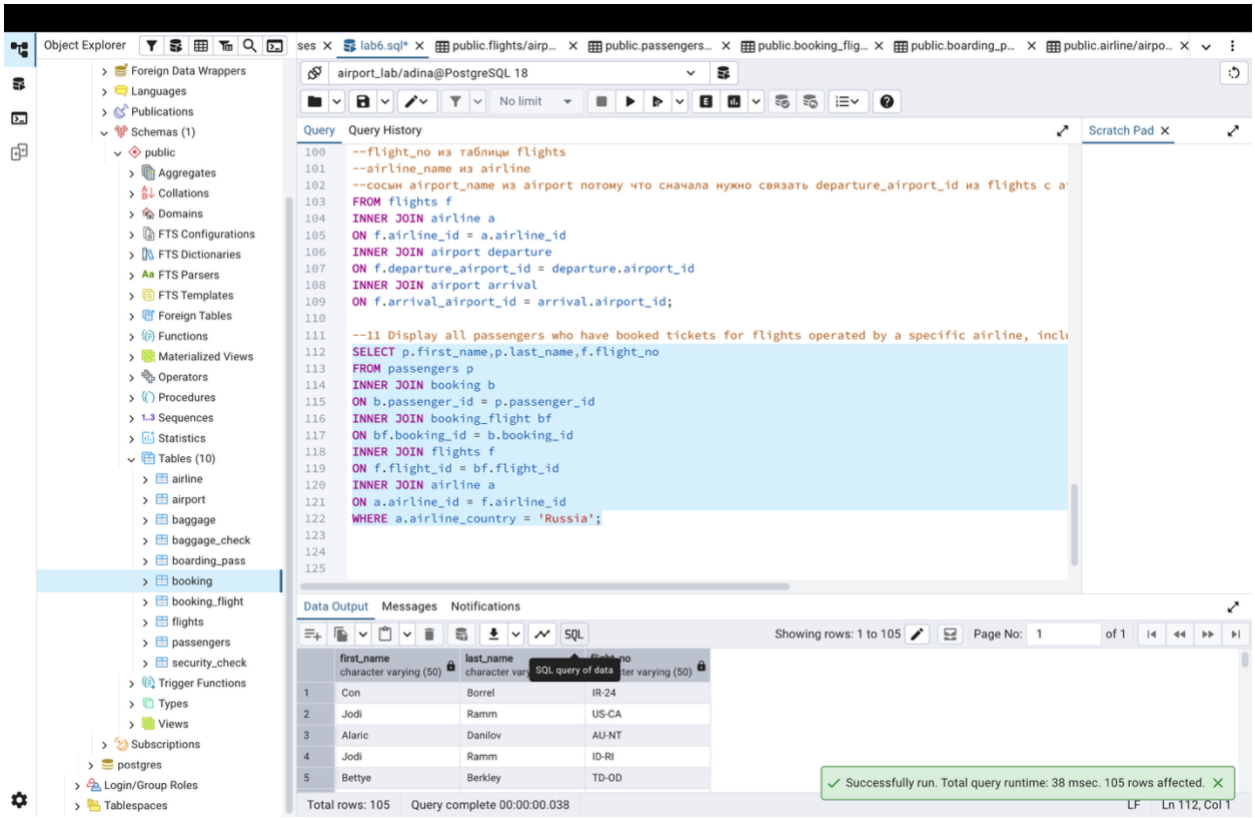


10. Write a query that lists each flight number, airline name, departure airport, and arrival airport by joining the relevant

tables.



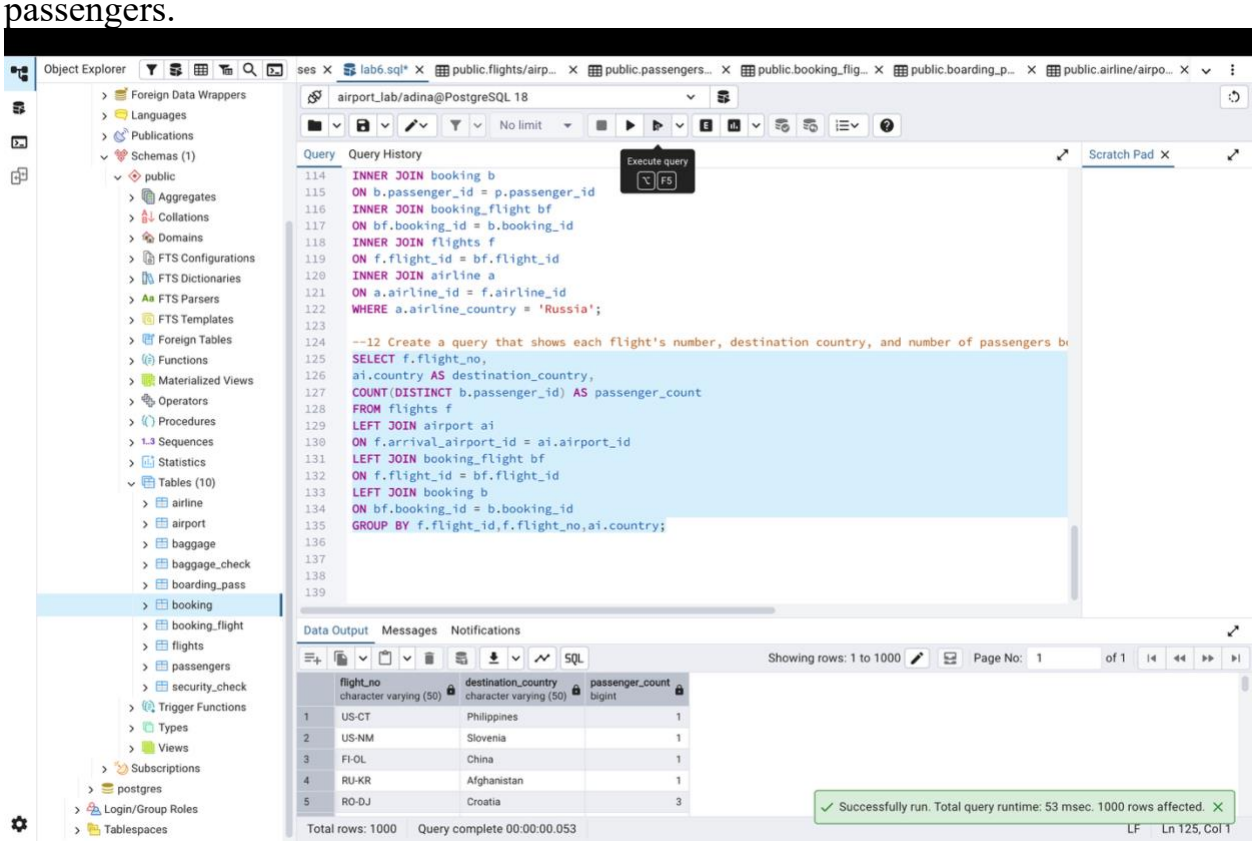
11. Display all passengers who have booked tickets for flights operated by a specific airline, including the passenger's full name and flight number.



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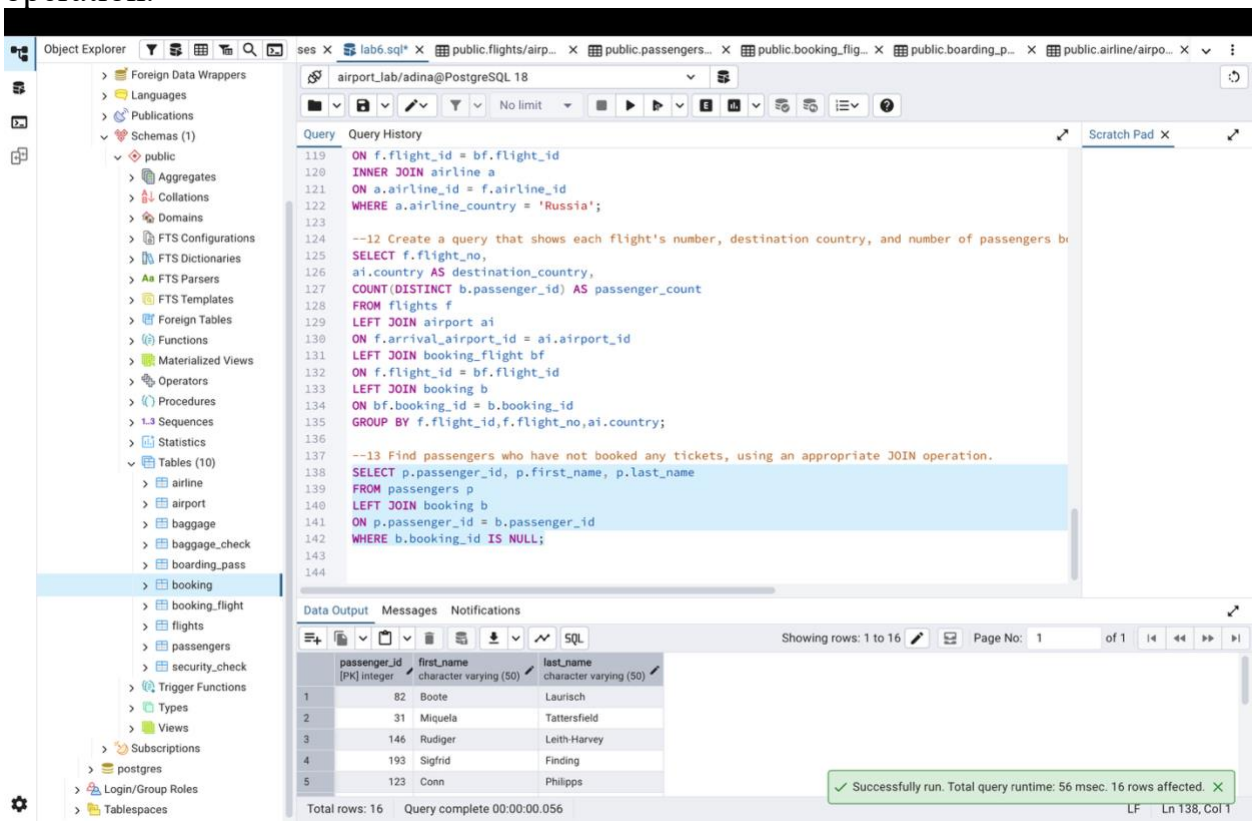
12. Create a query that shows each flight's number, destination country, and number of passengers booked on that flight. Include flights with no booked

passengers.



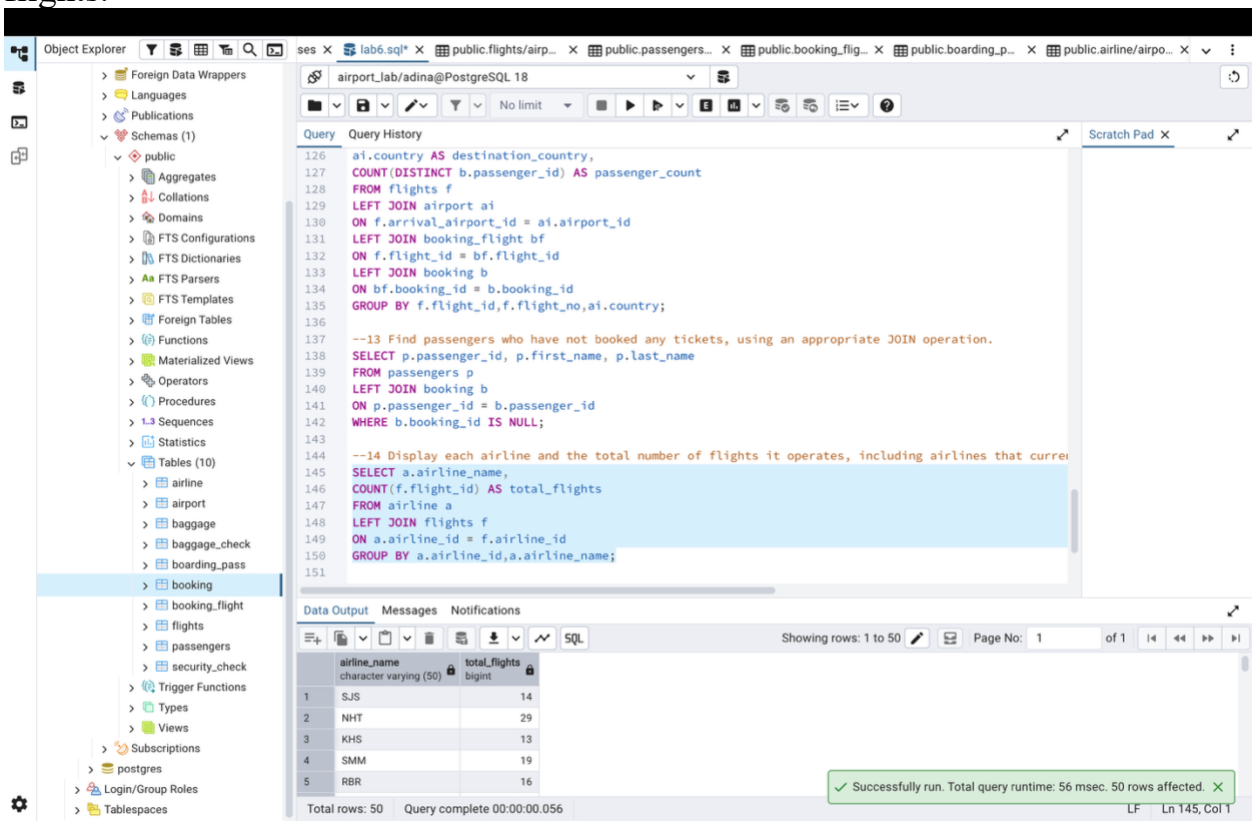
13.Find passengers who have not booked any tickets, using an appropriate JOIN

operation.



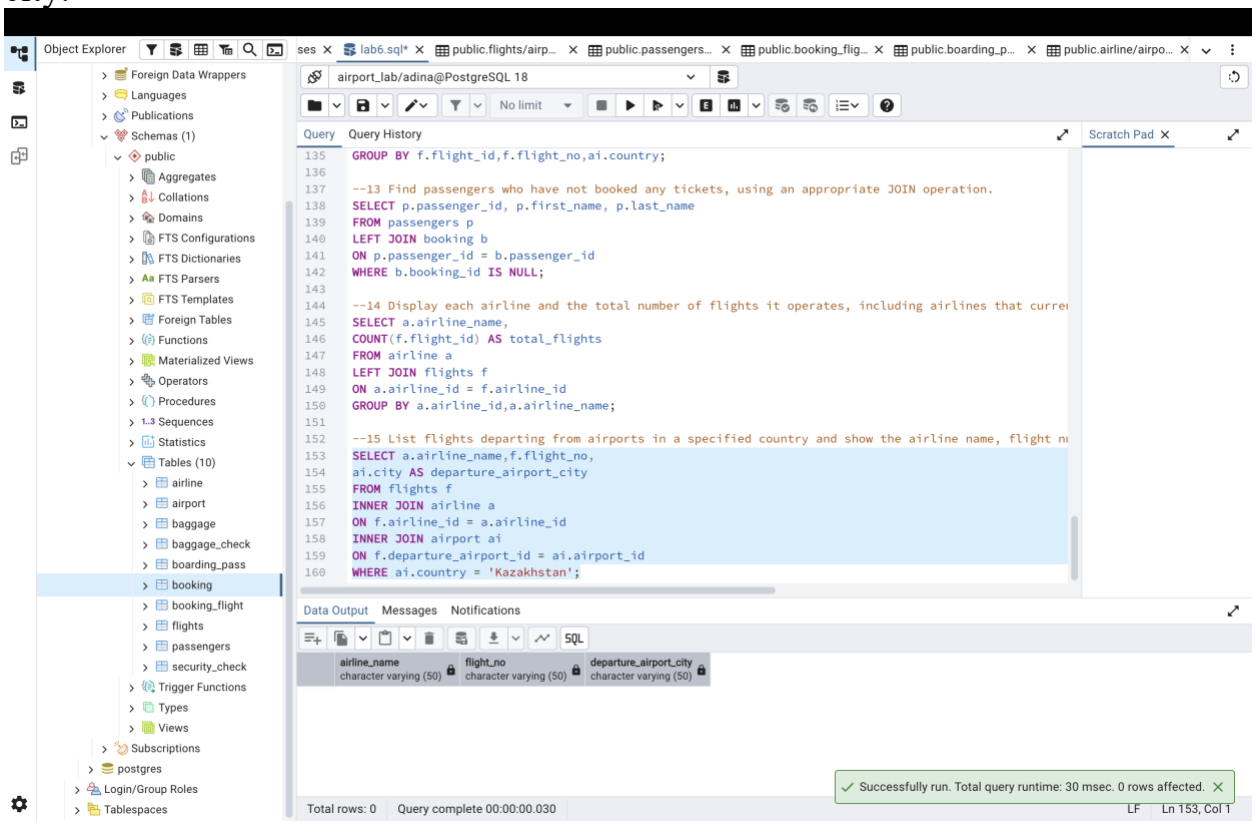
14. Display each airline and the total number of flights it operates, including airlines that currently have no

flights.



15. List flights departing from airports in a specified country and show the airline name, flight number, and departure airport

city.


Object Explorer
lab6.sql*
public.flights/airp...
public.passengers...
public.booking_flig...
public.boarding_p...
public.airline/airpo...
airport_lab/adina@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
GROUP BY f.flight_id,f.flight_no,ai.country;
--13 Find passengers who have not booked any tickets, using an appropriate JOIN operation.
SELECT p.passenger_id, p.first_name, p.last_name
FROM passengers p
LEFT JOIN booking b
ON p.passenger_id = b.passenger_id
WHERE b.booking_id IS NULL;
--14 Display each airline and the total number of flights it operates, including airlines that curre
SELECT a.airline_name,
COUNT(f.flight_id) AS total_flights
FROM airline a
LEFT JOIN flights f
ON a.airline_id = f.airline_id
GROUP BY a.airline_id,a.airline_name;
--15 List flights departing from airports in a specified country and show the airline name, flight n
SELECT a.airline_name,f.flight_no,
ai.city AS departure_airport_city
FROM flights f
INNER JOIN airline a
ON f.airline_id = a.airline_id
INNER JOIN airport ai
ON f.departure_airport_id = ai.airport_id
WHERE ai.country = 'Kazakhstan';
Data Output
Messages
Notifications
SQL
airline_name
character varying (50)
flight_no
character varying (50)
departure_airport_city
character varying (50)
Successfully run. Total query runtime: 30 msec. 0 rows affected.
Total rows: 0
Query complete 00:00:00.030
LF
Ln 153, Col 1
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
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